

A Provably Secure, MQ-Based W-OTS⁺

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Abstract

Hash-based signature (HBS) schemes provide a conservative foundation for post-quantum security, yet their formal proofs often rely on abstract hash properties that are difficult to verify in black-box heuristics. Recent research on SPHINCS⁺ has highlighted a significant gap in this regard: when concrete instantiations fail to strictly satisfy the idealized properties required by the proofs, the security of the scheme may be compromised. Such discrepancies motivate a shift toward instantiations anchored in structured algebraic problems, where the specific security properties required by HBS frameworks can be formally derived from well-studied computational hardness assumptions.

In this work, we present the first comprehensive instantiation of the W-OTS⁺ scheme based on multivariate quadratic (MQ) functions. We establish the *existential unforgeability under chosen message attack* (EU-CMA) of the scheme by providing a complete security reduction to the standard *MQ assumption* and the *one-wayness* of the MQ function family. Our analysis aligns with the rigorous standard-model requirements of the W-OTS⁺ framework: specifically, we prove that *second-preimage resistance* (SPR) inherits established average-case hardness while possessing worst-case **NP**-hardness, providing a complexity-theoretic guarantee that persists independently of specific algorithmic refinements. Furthermore, we bridge a critical gap in standard-model proofs by grounding the *undetectability* (UD) of the MQ family in its one-wayness, ensuring that the function's output distribution is theoretically indistinguishable from uniform. This construction provides a theoretically grounded alternative for HBS that prioritizes provable robustness derived from the inherent algebraic structure and well-studied hardness assumptions of the MQ problem.

Keywords: Multivariate Cryptography, Hash-Based Signatures, One-Time Signatures, Second-Preimage Resistance, Undetectability

1 Introduction

Hash-based signature (HBS) schemes have emerged as a cornerstone of post-quantum cryptography (PQC), offering a conservative and well-understood path toward quantum-resistant digital signatures. Their maturation is evidenced by the standardization of stateful schemes like XMSS [2, 3] and the selection of the stateless SPHINCS⁺ [6] as a NIST PQC standard. The primary allure of HBS lies in its *modular design and reductionist security*: the security of the entire scheme rests on a handful of well-defined properties of the underlying function family.

The Gap Between Idealized Proofs and Concrete Instantiations. Despite the elegance of HBS security reductions, a subtle yet critical gap persists between the abstract properties required by the proofs and the concrete hash functions used in practice. A striking example is the observation by Antonov [8] regarding the SPHINCS⁺ hash construction. The proof of SPHINCS⁺ [7] relies on the *distinct-function, multi-target second-preimage resistance* (DM-SPR) property; however, the standard keyed construction $\text{SHA-256}(\text{key} \parallel \text{message})$ fails to satisfy this property. This discrepancy was further exploited by Perlner et al. [9], who demonstrated a forgery attack that reduces the concrete security of SPHINCS⁺ by approximately 40 bits for certain parameters.

This line of research exposes a fundamental tension: while HBS proofs are rigorous, they often operate in *idealized hash models*. If the gap between the model and the instantiation is not bridged, the “provable security” of the scheme remains partially heuristic. This discrepancy motivates the pursuit of instantiations grounded in *explicit and structured hardness assumptions* from computational complexity theory, thereby transitioning beyond the “black-box” heuristics of generic hash functions.

W-OTS⁺ as a Flexible Foundation. A natural candidate for such an instantiation is W-OTS⁺ [10], the fundamental building block for both XMSS and SPHINCS⁺. The suitability of W-OTS⁺ for structured instantiations stems from a seminal technical shift: the removal of the requirement for *collision resistance* (CR) [10, 11]. Specifically, Hüsling established the EU-CMA security of W-OTS⁺ by reducing it to three properties of the underlying function family: *one-wayness* (OW), *second-preimage resistance* (SPR), and *undetectability* (UD) [10].

This shift is particularly advantageous for Multivariate Quadratic (MQ) functions. Historically, their inherent difficulty in achieving CR [18] has restricted their utility as general-purpose cryptographic hash functions. However, by leveraging the W-OTS⁺ framework, where CR is an unnecessary burden, we can harness the one-wayness of MQ systems—anchored in the **NP**-hardness of solving quadratic equations over finite fields [13]. This paper explores this intersection, asking: *Can the CR-free nature of W-OTS⁺ be leveraged to transform MQ functions into a more theoretically grounded and resilient foundation for post-quantum signatures?*

Our Contributions. In this work, we provide the first comprehensive instantiation and analysis of W-OTS⁺ based on MQ functions. By leveraging these algebraic primitives, we shift the security foundation of W-OTS⁺ from heuristic hash properties to structured and well-studied computational hardness assumptions. Our main technical contributions are as follows:

- **Complexity-Theoretic Analysis of SPR:** We provide a rigorous foundation for the SPR of the MQ function family. While we account for the average-case hardness by bridging our instantiation with results established in prior literature, our central theoretical contribution is a formal proof of the worst-case **NP**-hardness of finding second preimages for this family. This result ensures that the difficulty of finding second preimages persists even if specific average-case instances are subjected to future algorithmic refinements, providing a safeguard rooted in computational complexity.
- **Reductionist Proof of UD:** We provide a formal reduction showing that the UD of the MQ function family reduces to its OW. This ensures that under the MQ one-wayness assumption, the outputs of our construction are indistinguishable from uniform strings, fulfilling the stringent requirements of the W-OTS⁺ framework within the standard model while providing a concrete complexity-theoretic basis for this crucial property.
- **EU-CMA Security Reduction:** Leveraging the aforementioned properties, we present a complete instantiation of W-OTS⁺ and provide a rigorous security reduction. By grounding the scheme’s EU-CMA security in the standard MQ assumption and the MQ one-wayness assumption, we ensure that the construction’s safety is formally reducible to well-defined hardness guarantees rather than heuristic black-box functions.

Together, these contributions establish a formal bridge between multivariate cryptography and hash-based signatures, ensuring that the entire construction is supported by a seamless chain of rigorous complexity-theoretic proofs.

Organization. The remainder of this paper is organized as follows. Section 2 introduces the necessary preliminaries. Section 3 analyzes the cryptographic properties of the MQ function family. Section 4 presents the construction of MQ-based W-OTS⁺ and proves its EU-CMA security. Finally, Section 5 concludes the paper.

2 Preliminaries

We begin by recalling standard notation, following [12]. Let $x \xleftarrow{\$} X$ denote that x is sampled uniformly at random from a finite set X . Let $y \leftarrow \text{Alg}(x)$ denote the output of the algorithm Alg on input x . We write $\log$ for $\log_2$. The symbol $\oplus$ denotes bitwise exclusive-or, and $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and ceiling functions, respectively.

2.1 Digital Signature Schemes

Definition 1 (Digital Signature Schemes) Suppose $\mathcal{M}$ is a message space. A digital signature scheme is a triple of probabilistic polynomial-time algorithms $\text{Dss} = (\text{Kg}, \text{Sign}, \text{Vf})$ [12], defined as follows:

- $\text{Kg}(1^n)$ takes as input a security parameter 1^n and outputs a signing key sk and a verification key pk ;
- $\text{Sign}(\text{sk}, M)$ outputs a signature σ on a message $M \in \mathcal{M}$;
- $\text{Vf}(\text{pk}, \sigma, M)$ outputs 1 if σ is a valid signature on M under pk , and 0 otherwise.

The scheme is required to satisfy correctness, namely that for all $(\text{sk}, \text{pk}) \leftarrow \text{Kg}(1^n)$ and all $M \in \mathcal{M}$,

$$\text{Vf}(\text{pk}, \text{Sign}(\text{sk}, M), M) = 1.$$

The existential unforgeability under chosen-message attacks (EU-CMA) security of Dss is defined via the following experiment [12].

Experiment $\text{Exp}_{\text{Dss}(1^n)}^{\text{EU-CMA}}(\mathcal{A})$

- $(\text{sk}, \text{pk}) \leftarrow \text{Kg}(1^n)$;
- $(M^*, \sigma^*) \leftarrow \mathcal{A}^{\text{Sign}(\text{sk}, \cdot)}(\text{pk})$;
- Let $\{(M_i, \sigma_i)\}_{i=1}^{q_s}$ denote the query-response pairs corresponding to the adversary's signing oracle queries;
- Return 1 if and only if $\text{Vf}(\text{pk}, \sigma^*, M^*) = 1$ and $M^* \notin \{M_i\}_{i=1}^{q_s}$.

We define the success probability of $\mathcal{A}$ in the above experiment as

$$\text{Succ}_{\text{Dss}(1^n), q_s}^{\text{EU-CMA}}(\mathcal{A}) = \Pr[\text{Exp}_{\text{Dss}(1^n)}^{\text{EU-CMA}}(\mathcal{A}) = 1],$$

where the probability is taken over the randomness of Kg , Sign , and $\mathcal{A}$, and where q_s denotes the number of signing oracle queries made by $\mathcal{A}$.

Definition 2 (EU-CMA Security) We say that Dss is EU-CMA-secure if its insecurity is negligible in n , namely,

$$\text{InSec}^{\text{EU-CMA}}(\text{Dss}(1^n); t, q_s) := \sup_{\mathcal{A}} \text{Succ}_{\text{Dss}(1^n), q_s}^{\text{EU-CMA}}(\mathcal{A}) = \text{negl}(n),$$

where the supremum is taken over all probabilistic adversaries $\mathcal{A}$ running in time at most t and making at most q_s signing oracle queries.

2.2 Keyed Function Properties

From now on, we assume that

$$\mathcal{F}_n = \{f_k : \{0, 1\}^n \rightarrow \{0, 1\}^n\}_{k \in \mathcal{K}_n}$$

is a keyed function family, where $\mathcal{K}_n$ denotes the key space. The security of W-OTS^+ relies on the following properties of the underlying function family $\mathcal{F}_n$: one-wayness, second-preimage resistance, and undetectability (see Section 2.3.2).

Definition 3 (One-Wayness [12]) We say that $\mathcal{F}_n$ is one-way (OW) if for every probabilistic polynomial-time adversary $\mathcal{A}$,

$$\Pr[k \xleftarrow{\$} \mathcal{K}_n, x \xleftarrow{\$} \{0, 1\}^n, y = f_k(x), x' \leftarrow \mathcal{A}(k, y) : y = f_k(x')] = \text{negl}(n), \quad (1)$$

where the probability is taken over the randomness of k , x , and $\mathcal{A}$. Equivalently,

$$\text{InSec}^{\text{OW}}(\mathcal{F}_n; t) := \sup_{\mathcal{A}} \text{Succ}_{\mathcal{F}_n}^{\text{OW}}(\mathcal{A}) = \text{negl}(n),$$

where the supremum is taken over all probabilistic adversaries $\mathcal{A}$ running in time at most t , and where $\text{Succ}_{\mathcal{F}_n}^{\text{OW}}(\mathcal{A})$ denotes the probability in Equation (1).

Definition 4 (Second-Preimage Resistance [12]) We say that $\mathcal{F}_n$ is second-preimage resistant (SPR) if for every probabilistic polynomial-time adversary $\mathcal{A}$,

$$\Pr[k \xleftarrow{\$} \mathcal{K}_n, x \xleftarrow{\$} \{0, 1\}^n, x' \leftarrow \mathcal{A}(k, x) : f_k(x) = f_k(x') \wedge x \neq x'] = \text{negl}(n), \quad (2)$$

where the probability is taken over the randomness of k , x , and $\mathcal{A}$. Equivalently,

$$\text{InSec}^{\text{SPR}}(\mathcal{F}_n; t) := \sup_{\mathcal{A}} \text{Succ}_{\mathcal{F}_n}^{\text{SPR}}(\mathcal{A}) = \text{negl}(n),$$

where the supremum is taken over all probabilistic adversaries running in time at most t , and where $\text{Succ}_{\mathcal{F}_n}^{\text{SPR}}(\mathcal{A})$ denotes the probability in Equation (2).

To define undetectability, we first recall the notion of distinguishing advantage.

Definition 5 (Distinguishing Advantage [12]) Let $\mathcal{X}$ and $\mathcal{Y}$ be two distributions, and let an adversary $\mathcal{A}$ be given a sample drawn from one of them, chosen uniformly at random. The adversary outputs 1 if it believes the sample was drawn from $\mathcal{X}$ and 0 otherwise. The distinguishing advantage of $\mathcal{A}$ between $\mathcal{X}$ and $\mathcal{Y}$ is defined as

$$\text{Adv}_{\mathcal{X}, \mathcal{Y}}(\mathcal{A}) := |\Pr[\mathcal{A}(\mathcal{X}) = 1] - \Pr[\mathcal{A}(\mathcal{Y}) = 1]|,$$

where the probability is taken over the randomness of $\mathcal{A}$.

We consider the following two distributions over $\{0, 1\}^n \times \mathcal{K}_n$:

- $\mathcal{D}_{\text{UD}, \mathcal{U}}$: sample $k \xleftarrow{\$} \mathcal{K}_n$ and $u \xleftarrow{\$} \{0, 1\}^n$, and output (u, k) ;
- $\mathcal{D}_{\text{UD}, \mathcal{F}_n}$: sample $k \xleftarrow{\$} \mathcal{K}_n$ and $x \xleftarrow{\$} \{0, 1\}^n$, set $u = f_k(x)$, and output (u, k) .

The advantage of an adversary $\mathcal{A}$ against the undetectability of $\mathcal{F}_n$ is defined as

$$\text{Adv}_{\mathcal{F}_n}^{\text{UD}}(\mathcal{A}) := \text{Adv}_{\mathcal{D}_{\text{UD}, \mathcal{U}}, \mathcal{D}_{\text{UD}, \mathcal{F}_n}}(\mathcal{A}).$$

Definition 6 (Undetectability [12]) We say that $\mathcal{F}_n$ is undetectable (UD) if

$$\text{InSec}^{\text{UD}}(\mathcal{F}_n; t) := \sup_{\mathcal{A}} \text{Adv}_{\mathcal{F}_n}^{\text{UD}}(\mathcal{A}) = \text{negl}(n),$$

where the supremum is taken over all probabilistic polynomial-time adversaries $\mathcal{A}$ running in time at most t .

2.3 W-OTS⁺ Scheme and its Security

We briefly recall the W-OTS⁺ signature scheme and its security properties, following the presentation of Kudinov et al. [12], in order to make explicit the exact assumptions required of the underlying function family.

2.3.1 Construction

We now describe the W-OTS⁺ signature scheme [12].

Parameters. The parameters of the W-OTS⁺ scheme are positive integers defined as follows:

- n : the security parameter, which also determines the input and output length of the functions in $\mathcal{F}_n$;
- l_{msg} : the bit-length of messages to be signed;
- w : the Winternitz parameter, which determines the base- w representation and the compression level;
- l : the number of n -bit strings contained in a secret key, public key, and signature. It is computed as

$$l_1 = \left\lceil \frac{l_{\text{msg}}}{\log(w)} \right\rceil, \quad l_2 = \left\lceil \frac{\log(l_1(w-1))}{\log(w)} \right\rceil + 1, \quad l = l_1 + l_2.$$

Chaining Functions. The function family $\mathcal{F}_n$ is used to define the chaining functions. For $x \in \{0, 1\}^n$, a randomness vector $\mathbf{r} = (r_1, \dots, r_j) \in \{0, 1\}^{(n,j)}$, and integers $j \geq i \geq 0$, we define

$$c_k^i(x, \mathbf{r}) = \begin{cases} x, & \text{if } i = 0, \\ f_k(c_k^{i-1}(x, \mathbf{r}) \oplus r_i), & \text{if } i > 0. \end{cases}$$

Here, $\mathbf{r} \in \{0, 1\}^{(n,j)}$ denotes a sequence of j bit strings of length n .

Key Generation $\text{Kg}(1^n)$. First, sample a function key $k \xleftarrow{\$} \mathcal{K}_n$ and randomness $\mathbf{r} = (r_1, \dots, r_{w-1}) \xleftarrow{\$} \{0, 1\}^{(n,w-1)}$. Next, generate the secret key

$$\text{sk} = (\text{sk}_1, \dots, \text{sk}_l) \xleftarrow{\$} \{0, 1\}^{(n,l)}.$$

The public key is computed as

$$\text{pk} = (\text{pk}_0, \text{pk}_1, \dots, \text{pk}_l) = ((\mathbf{r}, k), c_k^{w-1}(\text{sk}_1, \mathbf{r}), \dots, c_k^{w-1}(\text{sk}_l, \mathbf{r})).$$

Signature Generation $\text{Sign}(\text{sk}, M)$. The signer first converts the message M into its base- w representation $M = (M_1, \dots, M_{l_1})$, where $M_i \in \{0, \dots, w-1\}$. The checksum is computed as

$$C = \sum_{i=1}^{l_1} (w-1-M_i),$$

and represented in base w as $C = (C_1, \dots, C_{l_2})$. Let $B = (b_1, \dots, b_l) = M \| C$. The signature on M is then computed as

$$\sigma = (\sigma_1, \dots, \sigma_l) = (c_k^{b_1}(\text{sk}_1, \mathbf{r}), \dots, c_k^{b_l}(\text{sk}_l, \mathbf{r})).$$

Signature Verification $\text{Vf}(\text{pk}, \sigma, M)$. The verifier first computes $B = (b_1, \dots, b_l)$ from M as above, and then checks whether

$$(\text{pk}_1, \dots, \text{pk}_l) = (c_k^{w-1-b_1}(\sigma_1, \mathbf{r}_{b_1+1, w-1}), \dots, c_k^{w-1-b_l}(\sigma_l, \mathbf{r}_{b_l+1, w-1})),$$

where

$$\mathbf{r}_{a,b} = \begin{cases} (r_a, r_{a+1}, \dots, r_b), & \text{if } b \geq a, \\ \varepsilon, & \text{otherwise.} \end{cases}$$

2.3.2 Security Theorem

When analyzing the EU-CMA security of W-OTS^+ , the adversary is allowed at most one signing query, i.e., $q_s = 1$. This restriction is inherent to the scheme, since an adversary obtaining two valid signature-query pairs can trivially forge a signature with probability 1.

Kudinov et al. [12] proved the following security bound.

Theorem 1 *Suppose $w, l_{\text{msg}} = \text{poly}(n)$. Let $\mathcal{F}_n = \{f_k : \{0, 1\}^n \rightarrow \{0, 1\}^n\}_{k \in \mathcal{K}_n}$ be a one-way, second-preimage resistant, and undetectable function family. Then the insecurity of the W-OTS^+ scheme against EU-CMA attacks is bounded by*

$$\begin{aligned} \text{InSec}^{\text{EU-CMA}}(\text{W-OTS}^+(1^n, w, l_{\text{msg}}); t, q_s = 1) < \\ lw \left(\text{InSec}^{\text{OW}}(\mathcal{F}_n; \tilde{t}) + w \cdot \text{InSec}^{\text{SPR}}(\mathcal{F}_n; \tilde{t}) + w \cdot \text{InSec}^{\text{UD}}(\mathcal{F}_n; \tilde{t}) \right), \end{aligned}$$

where $\tilde{t} = t + 3lw + w - 2$, and time is measured in the number of evaluations of functions from $\mathcal{F}_n$.

2.4 Another Solution Problem

To analyze the worst-case complexity of the second-preimage problem for the MQ function family, we recall some preliminaries on the *another solution problem* (ASP). The notion of ASP was first introduced by Ueda and Nagao [20] and concerns the task of finding a solution distinct from a given one for the same instance. A systematic complexity-theoretic treatment of ASP was later developed by Yato and Seta [21].

The definitions and results presented below follow the ASP framework of Yato and Seta [21].

Definition 7 Let $\Pi = (D, S, \sigma)$ be a function problem, where D denotes the set of instances, S is the set of all possible solutions, and for each $x \in D$, $\sigma(x)$ denotes the set of solutions for x . For a nonnegative integer n , the *n-another solution problem* of Π , denoted by $\Pi_{[n]} = (D_{[n]}, S, \sigma_{[n]})$, is defined as follows:

$$\begin{aligned} D_{[n]} &= \{(x, S_x) \mid S_x \subseteq \sigma(x), |S_x| = n\}, \\ \sigma_{[n]}(x, S_x) &= \sigma(x) \setminus S_x. \end{aligned}$$

The corresponding decision problem is called the *n-another solution decision problem* of Π .

To compare the complexity of another solution problems for different function problems, ASP reductions are defined as follows.

Definition 8 Let $\Pi_1 = (D_1, S_1, \sigma_1)$ and $\Pi_2 = (D_2, S_2, \sigma_2)$ be function problems. A pair of mappings $\varphi = (\varphi_D, \varphi_S)$ is called a *polynomial-time ASP reduction* from Π_1 to Π_2 if:

- φ_D is a polynomial-time computable mapping from D_1 to D_2 ;
- for every $x \in D_1$, φ_S is a polynomial-time computable bijection from $\sigma_1(x)$ to $\sigma_2(\varphi_D(x))$.

If such a reduction exists, we write $\Pi_1 \preceq_{\text{ASP}} \Pi_2$.

A fundamental property of ASP reductions is that they are preserved under the n -another solution transformation.

Proposition 2 Let Π and Π' be function problems. If $\Pi \preceq_{\text{ASP}} \Pi'$, then $\Pi_{[n]} \preceq_{\text{ASP}} \Pi'_{[n]}$ for any nonnegative integer n .

Similar to the class **NP** of decision problems, the class of *functional NP* problems, denoted by **FNP**, captures function problems whose solutions can be verified in polynomial time.

Definition 9 **FNP** is the class of function problems $\Pi = (D, S, \sigma)$ such that:

- there exists a polynomial p such that $|s| \leq p(|x|)$ for all $x \in D$ and all $s \in \sigma(x)$;
- for all $x \in D$ and $y \in S$, membership of y in $\sigma(x)$ can be decided in polynomial time.

ASP-completeness captures the intrinsic hardness of function problems under polynomial-time ASP reductions.

Definition 10 A function problem in **FNP** is called *ASP-complete* if every problem in **FNP** is polynomial-time ASP reducible to it.

Remark 1 ASP-completeness captures a notion of hardness that is particularly well suited to second-preimage problems. In such settings, an adversary is given one valid witness and is required to find a distinct one for the same instance. This focus on the structure of the solution space goes beyond mere existence of solutions, and aligns naturally with worst-case notions of second-preimage resistance in cryptographic applications.

ASP-completeness implies **NP**-completeness of the corresponding decision problem.

Theorem 3 For any ASP-complete function problem Π and any nonnegative integer n , the corresponding decision problem $\Pi_{[n]d}$ is **NP**-complete.

Yato and Seta further showed that satisfiability problems play a central role in the theory of ASP.

Theorem 4 ([21]) *Both SAT and 3SAT are ASP-complete.*

This follows from the ASP-completeness of SAT and the fact that $\text{SAT} \preceq_{\text{ASP}} 3\text{SAT}$.

In Section 3, we leverage this framework to establish the ASP-completeness of the MQ problem, thereby proving the NP-hardness of its second-preimage problem.

3 Cryptographic Properties of the MQ Function Family

We consider the following family of multivariate quadratic (MQ) functions:

$$\mathcal{H}_{n,m,q} := \left\{ f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m \left| \begin{array}{l} f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}}(\mathbf{x}) = (\mathbf{x}^\top \mathbf{A}_i \mathbf{x} + \mathbf{b}_i^\top \mathbf{x} + c_i)_{i=1}^m, \\ \text{where } \mathbf{A}_i \in \mathbb{F}_q^{n \times n} \text{ is upper triangular,} \\ \mathbf{b}_i \in \mathbb{F}_q^n, c_i \in \mathbb{F}_q, \quad i = 1, \dots, m. \end{array} \right. \right\}$$

Each function $f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}} \in \mathcal{H}_{n,m,q}$ is specified by

$$N_{n,m} = \left(\frac{1}{2}n(n+1) + n + 1 \right) m = \frac{1}{2}m(n+1)(n+2)$$

coefficients over $\mathbb{F}_q$. To match the security notions, we denote $k = (\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}) \in \mathbb{F}_q^{N_{n,m}} \triangleq \mathcal{K}$.

Remark 2 Throughout this section, the parameter n denotes the number of variables of the multivariate quadratic functions in $\mathcal{H}_{n,m,q}$, following standard conventions in multivariate cryptography. This should not be confused with the output length or the security parameter typically denoted by n in hash-based constructions such as W-OTS⁺.

In this section, we study the cryptographic properties of the MQ function family $\mathcal{H}_{n,m,q}$ that are required by the security framework of W-OTS⁺ as stated in Theorem 1. In particular, the security reduction of W-OTS⁺ relies on the standard notions of one-wayness (OW), second-preimage resistance (SPR), and undetectability (UD) of the underlying function family.

The remainder of this section is organized as follows. We first state the MQ-based assumptions supporting the one-wayness of $\mathcal{H}_{n,m,q}$. We then analyze its second-preimage resistance from both average-case and worst-case perspectives. Finally, we establish the undetectability of this family via a reduction from one-wayness.

3.1 MQ Assumptions and One-Wayness of $\mathcal{H}_{n,m,q}$

We begin by recalling the classical MQ problem.

Definition 11 (Multivariate Quadratic Problem, MQ [15]) The MQ problem is defined as follows.

Input: Positive integers n, m, q and a system of m quadratic polynomials $f_1, \dots, f_m$ over $\mathbb{F}_q$ in variables $(x_1, \dots, x_n)$.

Output: A vector $(u_1, \dots, u_n) \in \mathbb{F}_q^n$ such that $f_i(u_1, \dots, u_n) = 0$ for all $i = 1, \dots, m$.

The computational hardness of the MQ problem has been widely investigated in the literature. In particular, it is known that solving general systems of multivariate quadratic equations is **NP**-hard.

Theorem 5 ([13]) *The MQ problem is NP-hard.*

This worst-case hardness result indicates that the MQ problem captures computationally intractable structure from the perspective of classical complexity theory. Building on this foundation, cryptographic applications typically rely on additional assumptions concerning the hardness of solving MQ instances drawn from suitable random distributions.

Accordingly, we introduce two computational assumptions that will be used throughout this paper. The first concerns the hardness of solving random MQ systems, while the second captures the one-wayness of the corresponding multivariate quadratic function family.

Definition 12 (MQ Assumption) The (n, m, q) -MQ assumption states that it is computationally infeasible to solve the following problem: Given m upper triangular matrices $\mathbf{A}_1, \dots, \mathbf{A}_m \in \mathbb{F}_q^{n \times n}$ and m vectors $\mathbf{b}_1, \dots, \mathbf{b}_m \in \mathbb{F}_q^n$, all chosen independently and uniformly at random, find a non-zero $\mathbf{x} \in \mathbb{F}_q^n$ satisfying

$$\mathbf{x}^\top \mathbf{A}_i \mathbf{x} + \mathbf{b}_i^\top \mathbf{x} = 0 \quad \text{for all } i = 1, \dots, m.$$

Definition 13 (MQ One-Wayness Assumption) The (n, m, q) -MQ one-wayness assumption states that the MQ function family $\mathcal{H}_{n,m,q}$ is one-way.

Remark 3 The two assumptions stated above are formulated using slightly different representations of MQ systems, namely with and without constant terms. From a computational perspective, this distinction is inessential for the hardness of solving random MQ systems, as the dominant source of computational difficulty arises from the quadratic terms [15]. Accordingly, we freely adopt whichever formulation is more convenient for the purpose at hand when stating and using these assumptions in the subsequent average-case reductions and proofs.

The hardness of solving randomly generated MQ systems is well-supported by extensive cryptanalytic results, with the most computationally intensive instances occurring when $m \approx n$ [15, 22]. Specifically, for parameter choices where $m = \Theta(n)$, the best-known algorithms, such as Faugère’s F_4/F_5 or the XL algorithm, exhibit exponential complexity. Consequently, these assumptions are consistent with the

established security foundations of multivariate cryptography and provide a robust basis for instantiating the W-OTS⁺ framework using MQ-based primitives.

3.2 Second-Preimage Resistance of $\mathcal{H}_{n,m,q}$

In this subsection, we analyze the SPR property of the MQ function family $\mathcal{H}_{n,m,q}$ from both average-case and worst-case perspectives.

We first recall existing average-case results for SPR under the MQ assumption, which have been established for restricted families of MQ functions. We then show that these guarantees naturally extend to the full MQ function family considered in this work. Finally, we complement these average-case results with a complexity-theoretic worst-case hardness result for the corresponding second-preimage problem.

3.2.1 Average-Case Security

We first consider the following family of multivariate quadratic functions without constant terms:

$$\tilde{\mathcal{H}}_{n,m,q} := \left\{ f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m \mid f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}}(\mathbf{x}) = (\mathbf{x}^\top \mathbf{A}_i \mathbf{x} + \mathbf{b}_i^\top \mathbf{x})_{i=1}^m \right\},$$

where each $\mathbf{A}_i \in \mathbb{F}_q^{n \times n}$ is upper triangular and $\mathbf{b}_i \in \mathbb{F}_q^n$.

The average-case hardness of finding second preimages for $\tilde{\mathcal{H}}_{n,m,q}$ has been established under the MQ assumption.

Lemma 6 (MQ assumption $\Rightarrow$ SPR for $\tilde{\mathcal{H}}_{n,m,q}$ [19]) *Under the (n, m, q) -MQ assumption, the family $\tilde{\mathcal{H}}_{n,m,q}$ is second-preimage resistant.*

We now extend this result to the full MQ function family $\mathcal{H}_{n,m,q}$.

Theorem 7 (MQ assumption $\Rightarrow$ SPR for $\mathcal{H}_{n,m,q}$) *Under the (n, m, q) -MQ assumption, the family $\mathcal{H}_{n,m,q}$ is second-preimage resistant.*

Proof (sketch). The proof relies on the observation that adding a fixed output offset does not affect second-preimage resistance. Fix $(\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}})$ and suppose an adversary, given $\mathbf{x} \in \mathbb{F}_q^n$, finds $\mathbf{x}' \neq \mathbf{x}$ such that

$$f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}}(\mathbf{x}') = f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}}(\mathbf{x}).$$

Subtracting the constant vector $(c_i)_{i=1}^m$ from both sides yields

$$f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}}(\mathbf{x}') = f_{\vec{\mathbf{A}}, \vec{\mathbf{b}}}(\mathbf{x}),$$

which constitutes a valid second preimage for the function family $\tilde{\mathcal{H}}_{n,m,q}$. Hence, any adversary capable of finding second preimages for $\mathcal{H}_{n,m,q}$ can be directly transformed

into an adversary breaking the second-preimage resistance of $\tilde{\mathcal{H}}_{n,m,q}$. The claim then follows from the above lemma. $\square$

Hardness of average-case SPR. Based on this reduction, the average-case second-preimage resistance of $\mathcal{H}_{n,m,q}$ is inherited directly from the (n, m, q) -MQ assumption. In the standard parameter regime where $m = \Theta(n)$ and $q = \text{poly}(n)$, the average-case hardness of the (n, m, q) -MQ problem is widely recognized, thereby ensuring that $\mathcal{H}_{n,m,q}$ provides a robust level of SPR security in practice.

3.2.2 Worst-Case Hardness

To study the worst-case complexity of the second-preimage resistance (SPR) of $\mathcal{H}_{n,m,q}$, we consider the underlying function problem.

Definition 14 (Second-Preimage Problem for MQ) The second-preimage problem for MQ, denoted by MQ-SPP, is defined by the following input–output specification.

Input: Positive integers n, m, q , a vector $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{F}_q^n$, and m quadratic polynomials $f_1, \dots, f_m \in \mathbb{F}_q[x_1, \dots, x_n]$.

Output: A vector $\mathbf{u} = (u_1, \dots, u_n) \in \mathbb{F}_q^n$ such that $\mathbf{u} \neq \mathbf{y}$ and

$$f_i(\mathbf{u}) = f_i(\mathbf{y}) \quad \text{for all } i = 1, \dots, m.$$

We emphasize that worst-case hardness and average-case security are not equivalent in general. The results in this section should therefore be viewed as complementary to the average-case security guarantees discussed above. In particular, worst-case hardness rules out the existence of generic polynomial-time algorithms that break second-preimage resistance on all instances, and provides a complexity-theoretic foundation for the intrinsic difficulty of the problem.

We now relate MQ-SPP to the 1-another-solution version of MQ.

Proposition 8 *MQ-SPP and $\text{MQ}_{[1]}$ are polynomial-time equivalent.*

Proof. We construct polynomial-time reductions in both directions.

- $\text{MQ-SPP} \leq_P \text{MQ}_{[1]}$. Let $(f_1, \dots, f_m, \mathbf{y})$ be an instance of MQ-SPP. We define the polynomials

$$g_i(\mathbf{x}) := f_i(\mathbf{x}) - f_i(\mathbf{y}) \quad (i = 1, \dots, m).$$

Then $\mathbf{y}$ is a solution to the MQ instance $g_1(\mathbf{x}) = \dots = g_m(\mathbf{x}) = 0$. Any solution $\mathbf{u} \neq \mathbf{y}$ to this MQ system satisfies $f_i(\mathbf{u}) = f_i(\mathbf{y})$ for all i , and hence is a valid output for the original MQ-SPP instance. Conversely, any $\mathbf{u} \neq \mathbf{y}$ with $f_i(\mathbf{u}) = f_i(\mathbf{y})$ for all i is a distinct solution to $g_i(\mathbf{x}) = 0$ for all i . Therefore, solving the corresponding $\text{MQ}_{[1]}$ instance yields a solution to MQ-SPP.

- $\text{MQ}_{[1]} \leq_P \text{MQ-SPP}$. Let $((f_i)_{i=1}^m, \mathbf{y})$ be an instance of $\text{MQ}_{[1]}$, where $\mathbf{y}$ is promised to satisfy $f_i(\mathbf{y}) = 0$ for all i . Consider the MQ-SPP instance $(f_1, \dots, f_m, \mathbf{y})$. A vector $\mathbf{u} \neq \mathbf{y}$ is a valid output to this MQ-SPP instance if and only if $f_i(\mathbf{u}) = f_i(\mathbf{y}) = 0$ for all i , that is, if and only if it is another solution to the original MQ instance distinct from $\mathbf{y}$.

□

Hence, it suffices to analyze the hardness of $\text{MQ}_{[1]}$.

Theorem 9 *The MQ function problem is ASP-complete.*

Proof. Since 3SAT is ASP-complete (Theorem 4), it suffices to give a polynomial-time ASP reduction from 3SAT to MQ. We build on the classical construction of Patarin and Goubin [14] and present it in a form tailored to the ASP framework: we give the reductions and explicitly verify that each step induces a bijection between solution sets.

Step 1: arithmetization over $\mathbb{F}_2$ (from 3SAT to a cubic system). Let ψ be a 3SAT instance with variables $U = \{u_1, \dots, u_n\}$ and clauses $C = \{c_1, \dots, c_m\}$. Identify Boolean values with $\mathbb{F}_2 = \{0, 1\}$. Encode a literal ℓ by a polynomial $\text{val}(\ell) \in \mathbb{F}_2[x_1, \dots, x_n]$ via

$$\text{val}(u_i) = x_i, \quad \text{val}(\neg u_i) = 1 - x_i,$$

and encode disjunction by

$$\text{OR}(a, b) := a + b + ab.$$

For $a, b \in \{0, 1\}$ we have $\text{OR}(a, b) = a \vee b$. For each clause $c = (\ell_1 \vee \ell_2 \vee \ell_3)$, define the cubic polynomial

$$P_c(\mathbf{x}) := \text{OR}(\text{OR}(\text{val}(\ell_1), \text{val}(\ell_2)), \text{val}(\ell_3)).$$

Now form the system $(\mathcal{S})$ consisting of the m cubic equations

$$P_c(\mathbf{x}) = 1 \quad \text{for all } c \in C.$$

This construction is computable in polynomial time.

Bijection. Define $f_{\psi, (\mathcal{S})} : \sigma(\psi) \rightarrow \sigma(\mathcal{S})$ by mapping a satisfying assignment $a : U \rightarrow \{0, 1\}$ to

$$f_{\psi, (\mathcal{S})}(a) = (x_1, \dots, x_n) \in \mathbb{F}_2^n, \quad \text{where } x_i := a(u_i).$$

Conversely, define $g_{\psi, (\mathcal{S})} : \sigma(\mathcal{S}) \rightarrow \sigma(\psi)$ by

$$g_{\psi, (\mathcal{S})}(\mathbf{x}) = a_{\mathbf{x}}, \quad a_{\mathbf{x}}(u_i) := x_i.$$

For any $\mathbf{x} \in \mathbb{F}_2^n = \{0, 1\}^n$ and any clause $c = (\ell_1 \vee \ell_2 \vee \ell_3)$, the value $P_c(\mathbf{x})$ equals the truth value of c under $a_{\mathbf{x}}$ (by correctness of val and OR on $\{0, 1\}$). Hence $\mathbf{x} \in \sigma(\mathcal{S})$ implies $a_{\mathbf{x}} \in \sigma(\psi)$. Moreover, by construction we have $g_{\psi, (\mathcal{S})}(f_{\psi, (\mathcal{S})}(a)) = a$ and $f_{\psi, (\mathcal{S})}(g_{\psi, (\mathcal{S})}(\mathbf{x})) = \mathbf{x}$. Therefore, $f_{\psi, (\mathcal{S})}$ is a bijection.

Step 2: quadratization over $\mathbb{F}_2$ (from $(\mathcal{S})$ to an MQ system $(\mathcal{S}')$). Introduce auxiliary variables z_{ij} for $1 \leq i < j \leq n$, intended to represent $x_i x_j$. Replace every occurrence of $x_i x_j$ in $(\mathcal{S})$ by z_{ij} , and add the consistency constraints

$$z_{ij} = x_i x_j \quad (1 \leq i < j \leq n).$$

Denote the resulting quadratic system by $(\mathcal{S}')$. By construction, $(\mathcal{S}')$ is an MQ instance over $\mathbb{F}_2$ and is computable in polynomial time.

Bijection. Define $f_{(\mathcal{S}),(\mathcal{S}')} : \sigma(\mathcal{S}) \rightarrow \sigma(\mathcal{S}')$ by

$$f_{(\mathcal{S}),(\mathcal{S}')} (x_1, \dots, x_n) := ((x_i)_{i=1}^n, (z_{ij})_{1 \leq i < j \leq n}), \quad \text{where } z_{ij} := x_i x_j.$$

Define $g_{(\mathcal{S}),(\mathcal{S}')} : \sigma(\mathcal{S}') \rightarrow \sigma(\mathcal{S})$ as the projection

$$g_{(\mathcal{S}),(\mathcal{S}')} ((x_i), (z_{ij})) := (x_1, \dots, x_n).$$

If $((x_i), (z_{ij})) \in \sigma(\mathcal{S}')$, then the constraints $z_{ij} = x_i x_j$ force z_{ij} to equal the corresponding product, and substituting back yields that (x_i) satisfies the original cubic system $(\mathcal{S})$; hence $g_{(\mathcal{S}),(\mathcal{S}')}$ is well-defined. Moreover, $g_{(\mathcal{S}),(\mathcal{S}')} \circ f_{(\mathcal{S}),(\mathcal{S}')} = \text{id}$ on $\sigma(\mathcal{S})$, and $f_{(\mathcal{S}),(\mathcal{S}')} \circ g_{(\mathcal{S}),(\mathcal{S}')} = \text{id}$ on $\sigma(\mathcal{S}')$ (since z_{ij} is forced to be $x_i x_j$ in any solution of $(\mathcal{S}')$). Thus $f_{(\mathcal{S}),(\mathcal{S}')}$ is a bijection.

Step 3: lifting from $\mathbb{F}_2$ to an arbitrary field K (from $(\mathcal{S}')$ to $(\mathcal{S}'')$). Let $(\mathcal{S}')$ be an MQ instance over $\mathbb{F}_2$ in variables $\mathbf{v} = (v_1, \dots, v_N)$. We construct a quadratic system $(\mathcal{S}'')$ over K as follows.

(i) *Booleanity.* For each variable v_i , add the constraint

$$v_i(v_i - 1) = 0,$$

so any solution over K satisfies $v_i \in \{0, 1\} \subseteq K$.

(ii) *Multiplication.* Keep every product term ab unchanged; on Boolean inputs this agrees with multiplication in $\mathbb{F}_2$.

(iii) *Addition.* Replace each sum $a + b$ by

$$\text{Add}(a, b) := (a + b)(2 - (a + b)), \quad 2 := 1 + 1 \in K.$$

For $a, b \in \{0, 1\}$ we have $\text{Add}(a, b) = a \oplus b$. Applying this replacement recursively converts every equation of $(\mathcal{S}')$ into a quadratic equation over K . The resulting system is denoted by $(\mathcal{S}'')$ and is computable in polynomial time.

Bijection. Define $f_{(\mathcal{S}'),(\mathcal{S}'')} : \sigma(\mathcal{S}') \rightarrow \sigma(\mathcal{S}'')$ by viewing a Boolean solution $\mathbf{v} \in \{0, 1\}^N \subseteq \mathbb{F}_2^N$ as the same tuple in K^N under the embedding $\{0, 1\} \hookrightarrow K$. Conversely, define $g_{(\mathcal{S}'),(\mathcal{S}'')} : \sigma(\mathcal{S}'') \rightarrow \sigma(\mathcal{S}')$ by the same coordinate-wise reinterpretation: for $\mathbf{v} \in \sigma(\mathcal{S}'')$, the Booleanity constraints imply $\mathbf{v} \in \{0, 1\}^N$, so we regard $\mathbf{v}$ as an element of $\mathbb{F}_2^N$. On $\{0, 1\}$ -inputs, multiplication is preserved and $\text{Add}(\cdot, \cdot)$ agrees with $\mathbb{F}_2$ addition, hence $\mathbf{v}$ satisfies the original equations of $(\mathcal{S}')$ over $\mathbb{F}_2$. Therefore $g_{(\mathcal{S}'),(\mathcal{S}'')}$ is well-defined, and clearly $g_{(\mathcal{S}'),(\mathcal{S}'')} \circ f_{(\mathcal{S}'),(\mathcal{S}'')} = \text{id}$ and $f_{(\mathcal{S}'),(\mathcal{S}'')} \circ g_{(\mathcal{S}'),(\mathcal{S}'')} = \text{id}$ on the respective solution sets. Thus $f_{(\mathcal{S}'),(\mathcal{S}'')}$ is a bijection.

Composing Steps 1–3 yields a polynomial-time ASP reduction from 3SAT to MQ. Since 3SAT is ASP-complete, the MQ function problem is ASP-complete. $\square$

Combining Theorem 3 and Theorem 9, we obtain:

Corollary 10 *The decision version of $\text{MQ}_{[1]}$ is $\mathbf{NP}$ -complete.*

Since Proposition 8 gives a polynomial-time equivalence between MQ-SPP and $\text{MQ}_{[1]}$, we conclude:

Theorem 11 *The decision version of MQ-SPP is $\mathbf{NP}$ -complete.*

As shown above, the search version of MQ-SPP corresponds exactly to finding second preimages for the MQ function family. Therefore, the decision hardness of MQ-SPP implies worst-case hardness of solving the second-preimage problem for the MQ family.

Theorem 12 *Breaking the worst-case second-preimage resistance of the MQ function family $\mathcal{H}_{n,m,q}$ is $\mathbf{NP}$ -hard.*

The establishment of the worst-case $\mathbf{NP}$ -hardness of MQ-SPP provides a robust complexity-theoretic foundation for our construction. While cryptographic deployments primarily rely on average-case hardness assumptions for practical security parameters, the $\mathbf{NP}$ -hardness result serves as a critical "safety net." It ensures that the task of finding second preimages for the MQ function family is inherently as difficult as solving the most challenging problems within the class $\mathbf{NP}$. Consequently, even if future algorithmic refinements were to impact specific average-case instances, the underlying structure of the second-preimage problem maintains a formal barrier rooted in computational complexity theory, shielding the scheme's security against unforeseen gaps in heuristic difficulty.

3.3 Undetectability of $\mathcal{H}_{n,m,q}$

In this subsection, we establish the computational hardness of the undetectability property of $\mathcal{H}_{n,m,q}$ by presenting a reduction from undetectability to one-wayness. Specifically, we show that any efficient distinguisher for the undetectability game can be transformed into an efficient algorithm that inverts $\mathcal{H}_{n,m,q}$ on a random instance.

Theorem 13 ($\text{OW} \Rightarrow \text{UD}$ for $\mathcal{H}_{n,m,q}$) *Let $q = \text{poly}(n)$. Suppose there exists a probabilistic algorithm $\mathcal{D}$ running in time t that distinguishes the distributions $\mathcal{D}_{\text{UD}, \mathcal{H}_{n,m,q}}$ and $\mathcal{D}_{\text{UD}, \mathcal{U}}$ with advantage ϵ . Then there exists a probabilistic algorithm $\mathcal{A}^{\mathcal{D}}$ that, given a random one-wayness challenge $(k, \mathbf{y})$ with $\mathbf{y} = f_k(\mathbf{x})$, recovers $\mathbf{x}$ with probability $1 - o(1)$ and running time $t' = \tilde{O}(t \cdot nq/\epsilon^2)$.*

Proof. Our proof draws upon certain ideas from the hardness proofs of the Learning With Errors (LWE) problem [23].

Without loss of generality, we assume that $\Pr[\mathcal{D}(\mathcal{D}_{\text{UD}, \mathcal{U}}) = 1] = \frac{1}{2} - \frac{\epsilon}{2}$, and hence $\Pr[\mathcal{D}(\mathcal{D}_{\text{UD}, \mathcal{H}_{n,m,q}}) = 1] = \frac{1}{2} + \frac{\epsilon}{2}$.

Suppose $\mathcal{A}^D$ receives a key-image pair $(k, \mathbf{y})$, where k is drawn uniformly from $\mathcal{K} = \mathbb{F}_q^{N_{n,m}}$ and $\mathbf{y} = f_k(\mathbf{x})$, with $\mathbf{x} = (x_1, \dots, x_n)$ sampled uniformly from $\mathbb{F}_q^n$. The algorithm $\mathcal{A}^D$ is designed to guess each coordinate of the secret $\mathbf{x}$. We describe $\mathcal{A}^D$ in Algorithm 1, where we denote $\mathbb{F}_q = \{\alpha_1, \dots, \alpha_q\}$.

Algorithm 1 $\mathcal{A}^D$

Input: a one-wayness challenge $(k, \mathbf{y})$
Output: $\mathbf{x}$ such that $f_k(\mathbf{x}) = \mathbf{y}$

- 1: Parse k as $(\mathbf{A}_p, \mathbf{b}_p, c_p)_{p=1}^m$, where $\mathbf{A}_p \in \mathbb{F}_q^{n \times n}$, $\mathbf{b}_p \in \mathbb{F}_q^n$, and $c_p \in \mathbb{F}_q$
- 2: Set the repetition parameter $L = \tilde{O}(1/\epsilon^2)$
- 3: **for** $i = 1, 2, \dots, n$ **do**
- 4: **for** $j = 1, 2, \dots, q$ **do**
- 5: Set $g \leftarrow \alpha_j$
- 6: **for** $\ell = 1, 2, \dots, L$ **do**
- 7: Choose a random mask $t = (t_1, \dots, t_m) \xleftarrow{\$} \mathbb{F}_q^m$
- 8: **for** $p = 1, 2, \dots, m$ **do**
- 9: $\mathbf{b}'_p \leftarrow \mathbf{b}_p + t_p \mathbf{e}_i$, where $\mathbf{e}_i$ is the i -th unit vector in $\mathbb{F}_q^n$
- 10: $\mathbf{A}'_p \leftarrow \mathbf{A}_p$
- 11: $c'_p \leftarrow c_p$
- 12: $y'_p \leftarrow y_p + g t_p$
- 13: **end for**
- 14: Define $k' \leftarrow (\mathbf{A}'_p, \mathbf{b}'_p, c'_p)_{p=1}^m$
- 15: Define $\mathbf{y}' \leftarrow (y'_1, \dots, y'_m)$
- 16: Let $d_\ell \leftarrow D(k', \mathbf{y}')$
- 17: **end for**
- 18: **if** $\text{maj}(d_1, \dots, d_L) = 1$ **then**
- 19: Set $x_i \leftarrow g$
- 20: **break**
- 21: **end if**
- 22: **end for**
- 23: **end for**
- 24: Output $\mathbf{x} = (x_1, \dots, x_n)$

At a high level, the adversary $\mathcal{A}^D$ recovers the secret vector $\mathbf{x}$ in a coordinate-wise manner. For a fixed coordinate i and a candidate value $g \in \mathbb{F}_q$, the algorithm constructs modified instances $(k', \mathbf{y}')$ by injecting a random mask into the i -th linear coefficient of the MQ system and adjusting the output accordingly. This transformation ensures that when $g = x_i$, the resulting pair $(k', \mathbf{y}')$ is distributed according to the “real” distribution $\mathcal{D}_{\text{UD}, \mathcal{H}_{n,m,q}}$, whereas when $g \neq x_i$, the distribution of $(k', \mathbf{y}')$ is identical to the “uniform” distribution $\mathcal{D}_{\text{UD}, \mathcal{U}}$. By repeatedly querying the distinguisher D on these instances and applying a majority vote, $\mathcal{A}^D$ can reliably identify

the correct value of each coordinate. The algorithm proceeds coordinate by coordinate, assigning the value of x_i to the first candidate g that passes the majority test and then stopping.

We now analyze the behavior of $\mathcal{A}^D$. Fix any coordinate $i \in \{1, \dots, n\}$ and any candidate value $g \in \mathbb{F}_q$. In each repetition $\ell \in \{1, \dots, L\}$, the reduction samples a fresh mask $t = (t_1, \dots, t_m) \xleftarrow{\$} \mathbb{F}_q^m$ and defines, for every $p \in \{1, \dots, m\}$,

$$\mathbf{b}'_p := \mathbf{b}_p + t_p \mathbf{e}_i, \quad \mathbf{A}'_p := \mathbf{A}_p, \quad c'_p := c_p, \quad y'_p := y_p + g t_p,$$

where $\mathbf{e}_i$ is the i -th unit vector in $\mathbb{F}_q^n$. Let $k' := (\mathbf{A}'_p, \mathbf{b}'_p, c'_p)_{p=1}^m$ and $y' := (y'_1, \dots, y'_m)$. Writing the original challenge as $y_p = \mathbf{x}^\top \mathbf{A}_p \mathbf{x} + \mathbf{b}_p^\top \mathbf{x} + c_p$, we obtain

$$y'_p = y_p + g t_p = \mathbf{x}^\top \mathbf{A}_p \mathbf{x} + \mathbf{b}_p^\top \mathbf{x} + c_p + g t_p = \mathbf{x}^\top \mathbf{A}'_p \mathbf{x} + (\mathbf{b}'_p)^\top \mathbf{x} + c'_p + (g - x_i) t_p.$$

This leads to the following distributional dichotomy.

(i) The case $g = x_i$. Then $(g - x_i)t_p = 0$ for all p , and thus $y'_p = \mathbf{x}^\top \mathbf{A}'_p \mathbf{x} + (\mathbf{b}'_p)^\top \mathbf{x} + c'_p$. Equivalently, $\mathbf{y}' = f_{k'}(\mathbf{x})$. Therefore, the pair $(k', \mathbf{y}')$ is distributed exactly as in the “real” distribution $\mathcal{D}_{\text{UD}, \mathcal{H}_{n,m,q}}$ (namely, sample k' uniformly and output $(k', f_{k'}(\mathbf{x}))$ for uniform $\mathbf{x}$).

(ii) The case $g \neq x_i$. Since $t_p \xleftarrow{\$} \mathbb{F}_q$ and $g - x_i \neq 0$, the product $(g - x_i)t_p$ is uniformly distributed over $\mathbb{F}_q$. Consequently, each coordinate y'_p is uniform over $\mathbb{F}_q$ and independent of k' , and hence $(k', \mathbf{y}')$ is distributed identically to the “uniform” distribution $\mathcal{D}_{\text{UD}, \mathcal{U}}$ (namely, output (k', u) for uniform $u \leftarrow \mathbb{F}_q^m$).

For fixed (i, g) , define Bernoulli random variables $d_\ell := \mathcal{D}(k', y') \in \{0, 1\}$ over independent repetitions $\ell = 1, \dots, L$, and let $S := \sum_{\ell=1}^L d_\ell$. By the above dichotomy,

$$\Pr[d_\ell = 1 \mid g = x_i] = \frac{1}{2} + \frac{\epsilon}{2} \quad \text{and} \quad \Pr[d_\ell = 1 \mid g \neq x_i] = \frac{1}{2} - \frac{\epsilon}{2}.$$

We decide “ $g = x_i$ ” iff $\text{maj}(d_1, \dots, d_L) = 1$, i.e., iff $S \geq L/2$.

Using Hoeffding’s inequality, we have

$$\Pr\left[S < \frac{L}{2} \mid g = x_i\right] = \Pr\left[S - \mathbb{E}[S] \leq -\frac{\epsilon}{2}L\right] \leq \exp\left(-\frac{\epsilon^2}{2}L\right),$$

and similarly

$$\Pr\left[S \geq \frac{L}{2} \mid g \neq x_i\right] = \Pr\left[S - \mathbb{E}[S] \geq \frac{\epsilon}{2}L\right] \leq \exp\left(-\frac{\epsilon^2}{2}L\right).$$

Choosing

$$L = \tilde{O}\left(\frac{1}{\epsilon^2}\right)$$

makes the error probability for each fixed pair (i, g) at most $(nq)^{-c}$ for an arbitrary constant $c > 0$ (by adjusting the hidden constant in $\tilde{O}(\cdot)$). Applying a union bound over all nq tests, the probability that any test errs is at most

$$(nq) \cdot (nq)^{-c} = (nq)^{1-c},$$

which is $o(1)$ for $c > 1$. In particular, with overwhelming probability, for each coordinate i there exists a unique candidate g that passes the majority test. Accordingly, $\mathcal{A}^D$ recovers all coordinates of $\mathbf{x}$ with probability $1 - o(1)$.

Finally, since $q = \text{poly}(n)$, and $\mathcal{A}^D$ makes nqL invocations of $\mathcal{D}$, so its running time is

$$t' = \tilde{O}\left(\frac{t \cdot nq}{\epsilon^2}\right),$$

where $\tilde{O}(\cdot)$ hides polylogarithmic factors in n . This completes the proof. $\square$

Hardness and Theoretical Significance of Undetectability. In the standard parameter regime where $m = \Theta(n)$ and $q = \text{poly}(n)$, the (n, m, q) -MQ one-wayness assumption is widely recognized as a robust computational foundation. Our reduction demonstrates that for $\mathcal{H}_{n,m,q}$, undetectability is not an independent heuristic or an idealized black-box property, but a provable consequence of its one-wayness. Consequently, in this natural setting, the computational hardness of the UD property is formally reducible to the (n, m, q) -MQ one-wayness assumption, providing a theoretically grounded security model where the internal indistinguishability of the mappings is rigorously anchored in the foundational task of inverting them.

4 The MQ-based W-OTS⁺ Scheme

This section presents an instantiation of the W-OTS⁺ scheme using MQ functions, along with its provable security guarantees.

In the W-OTS⁺ framework, the underlying function family $\mathcal{F}_n = \{f_k : \{0, 1\}^n \rightarrow \{0, 1\}^n\}$ consists of length-preserving functions, i.e., the input and output lengths are both n bits. We instantiate $\mathcal{F}_n$ using an MQ function family $\mathcal{H}_{m,v,q}$, where v denotes the number of variables and m the number of quadratic equations.

In our instantiation, we adopt the natural square setting $m = v$, so that each function $f_k \in \mathcal{H}_{v,v,q}$ defines a map $\mathbb{F}_q^v \rightarrow \mathbb{F}_q^v$. Each output vector in $\mathbb{F}_q^v$ is encoded as a bit string of length

$$n = v \cdot \lceil \log_2 q \rceil,$$

thereby matching the length-preserving requirement of the W-OTS⁺ construction. While other parameter choices could in principle be accommodated via padding or compression, we focus on the square setting $m = v$, in which the MQ function naturally defines a map from $\mathbb{F}_q^v$ to itself. This choice yields a conceptually simple and direct length-preserving instantiation that fits seamlessly into the W-OTS⁺ framework.

Definition 15 (MQ-based W-OTS⁺) The MQ-based W-OTS⁺ scheme (MQ-W-OTS⁺) is the W-OTS⁺ signature scheme in which the underlying length-preserving function family $\mathcal{F}_n = \{f_k : \{0, 1\}^n \rightarrow \{0, 1\}^n\}$ is instantiated by the multivariate quadratic function family

$\mathcal{H}_{v,v,q}$ as described above, with the natural square setting $m = v$ and the canonical bit encoding $n = v \cdot \lceil \log_2 q \rceil$.

Now we introduce the EU-CMA security theorem of the MQ-based W-OTS⁺ scheme.

Theorem 14 (EU-CMA security of MQ-W-OTS⁺) *Let $q, w, l_{\text{msg}} = \text{poly}(n)$ and $v = n / \lceil \log_2 q \rceil$.*

Under the (v, v, q) -MQ assumption and the (v, v, q) -MQ one-wayness assumption, the MQ-based W-OTS⁺ scheme is EU-CMA secure. Specifically, the insecurity of the scheme against a single signing query is negligible in the security parameter n :

$$\text{InSec}^{\text{EU-CMA}}(\text{MQ-W-OTS}^+(1^n, w, l_{\text{msg}}); t, q_s = 1) = \text{negl}(n).$$

Proof (sketch). The security of MQ-W-OTS⁺ follows by applying the cryptographic attributes of the MQ family established in Section 3 to the general W-OTS⁺ framework. According to Theorem 1, the EU-CMA security of W-OTS⁺ in the standard model reduces to the OW, SPR, and UD of the underlying function family. For our instantiation $\mathcal{H}_{v,v,q}$, these properties are established as follows:

- **OW:** The one-wayness of $\mathcal{H}_{v,v,q}$ is directly specified by the (v, v, q) -MQ one-wayness assumption.
- **SPR:** Its average-case resistance is guaranteed by the (v, v, q) -MQ assumption (Section 3.2), with additional support from the **NP**-hardness of MQ-SPP in the worst case.
- **UD:** This property is formally reduced to the (v, v, q) -MQ one-wayness assumption in Section 3.3, ensuring the output indistinguishability in the standard model.

Substituting these reductionist links into Theorem 1 ensures that any efficient EU-CMA adversary against MQ-W-OTS⁺ can be transformed into an algorithm that violates either the (v, v, q) -MQ one-wayness assumption or the (v, v, q) -MQ assumption, thus completing the proof. $\square$

5 Conclusion

In this paper, we harness the MQ function family to instantiate the W-OTS⁺ scheme, yielding the first provably secure W-OTS⁺ based on explicit and well-studied computational hardness assumptions.

Beyond the immediate utility in digital signatures, our results suggest that the cryptographic potential of MQ functions remains to be fully explored. Such explorations may lead to further provably secure instantiations of hash-based primitives based on both standard and structured computational hardness assumptions.

Declarations

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